

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built full-stack workflow automation around structured forms, public-site data lookups, browser actions, backend state, and operator-visible failure recovery.
- Implemented JavaScript/Node.js services that generated validated AI-assisted drafts before handing them into filing automation, cutting manual effort by **85%**.
- Packaged and deployed services with Docker, Google Cloud Run, and GitHub Actions, improving release speed by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python tooling for MRI segmentation workflows, connecting preprocessing, model inference, generated masks, and 3D visualization in an existing desktop application.
- Improved runtime by **87%** with asynchronous and parallel execution over large medical-image volumes.
- Debugged cross-module integration issues across Python data handling, NumPy, PyTorch/ONNX-style inference, and VTK/wxPython UI behavior.

Open Source

Contributions 

- Contributed maintainer-reviewed patches across **Tiled**, **Kuadrant/Kyverno**, VideoLAN, CRIU, and IOOS, spanning UI/compatibility fixes, backend integration behavior, parser robustness, and Zip Slip path validation.

Projects

Hyoka | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, Next.js, TypeScript, Docker*

- Built a **reliability control plane for AI agents** with trace/event ingestion, authenticated APIs, suites, candidates, validation runs, gate decisions, promotions, audit logs, and a Next.js operations dashboard.
- Implemented SQLAlchemy/Alembic metadata models, project-scoped API keys, content-addressed artifacts, signed manifests, worker-owned execution leases, and Docker Compose deployment with Postgres-backed services.

Seaweed: Contest Platform | *TypeScript, React, Next.js, Go, PostgreSQL, Judge0, Firebase*

GitHub 

- Built a **React/Next.js assessment UI** with auth-gated problem pages, code editor, submissions, admin review/shortlisting, and live ranked leaderboards.
- Implemented Go judging APIs with Judge0, PostgreSQL rankings/materialized views, Redis-backed runtime state, and Docker/Kubernetes deployment for **500+ concurrent users** and **sub-5s P95 verdict latency**.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS*

GitHub 

- Built backend workflows that coordinate content transformation, image storage, autosave, durable job status, and browser automation across external publishing platforms.
- Used RabbitMQ, Redis, PostgreSQL, and Dockerized Go services to keep long-running integrations recoverable, auditable, and simple to operate.

Technical Skills

Languages: Python, JavaScript, TypeScript, Go, SQL

Frameworks: FastAPI, React, Next.js, Node.js, Express, Echo, PyTorch

Tools: PostgreSQL, MySQL, SQLite, Redis, Docker, AWS, GCP, GitHub Actions, Postman, Firebase